VICTORY 2027

LIGHT

LECTURE-01 PHYSICS

Shailendra sir



Topics To Be Covered

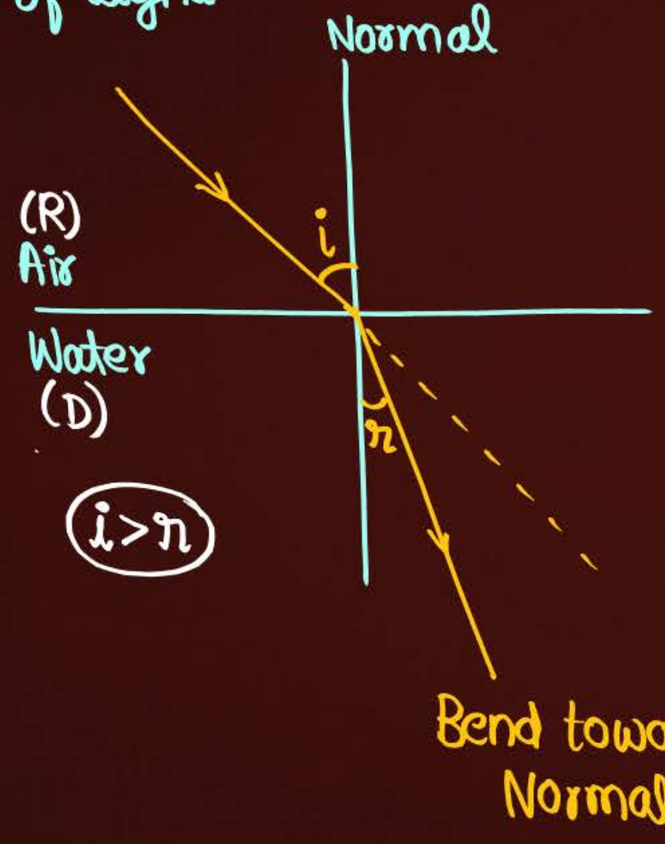
1. REVISION
2. REFRACTION OF LIGHT
3. LATERAL DISPLACEMENT
4. QUESTIONS

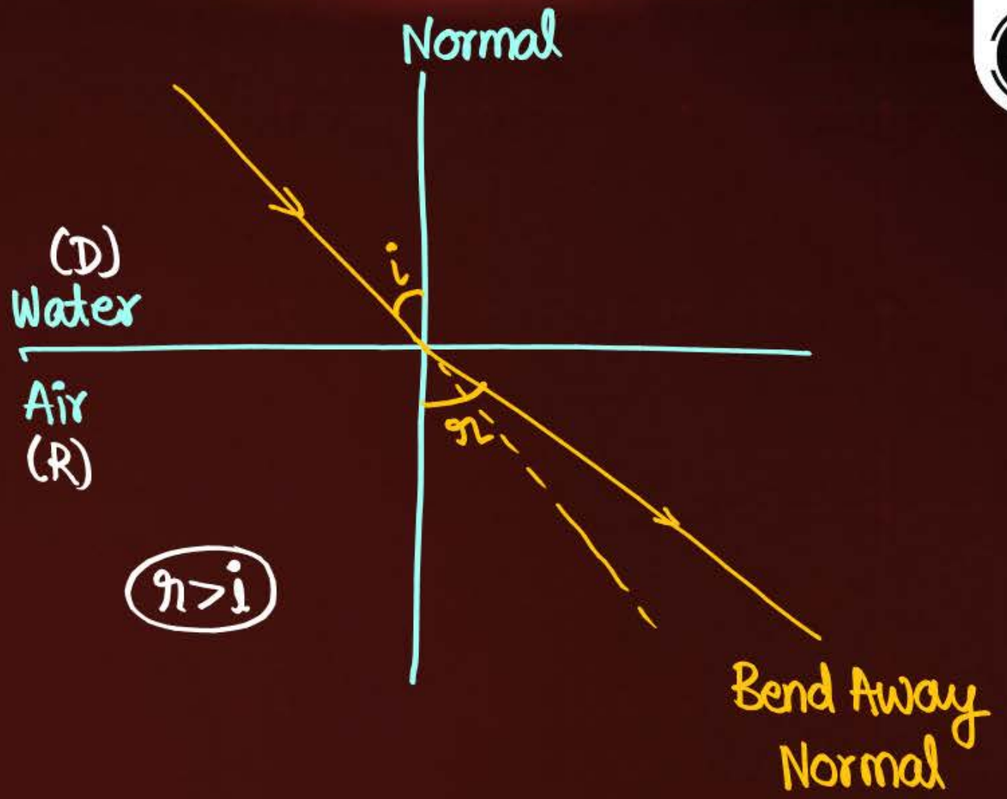

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REVISION

Differents Medium → Different Speeds of light.

Refraction of light





Medium

- Rarer Medium: $v \uparrow \; n \downarrow$
  - eg. Air, Water, Vacuum
- Denser Medium: $v \downarrow \; n \uparrow$
  - Water, Glass, Air

Refractive Index of Medium

Represents speed of light in medium.

$$\mu \text{ or } n = \frac{c}{v} \Rightarrow n \propto \frac{1}{v}$$

$$\mu \text{ or } n = \frac{3 \times 10^8}{v}$$

For Water $\mu = \frac{4}{3}$

$$\mu = \frac{c}{v}$$

$$\frac{4}{3} = \frac{c}{v}$$

$$V_w = \frac{3c}{4}$$

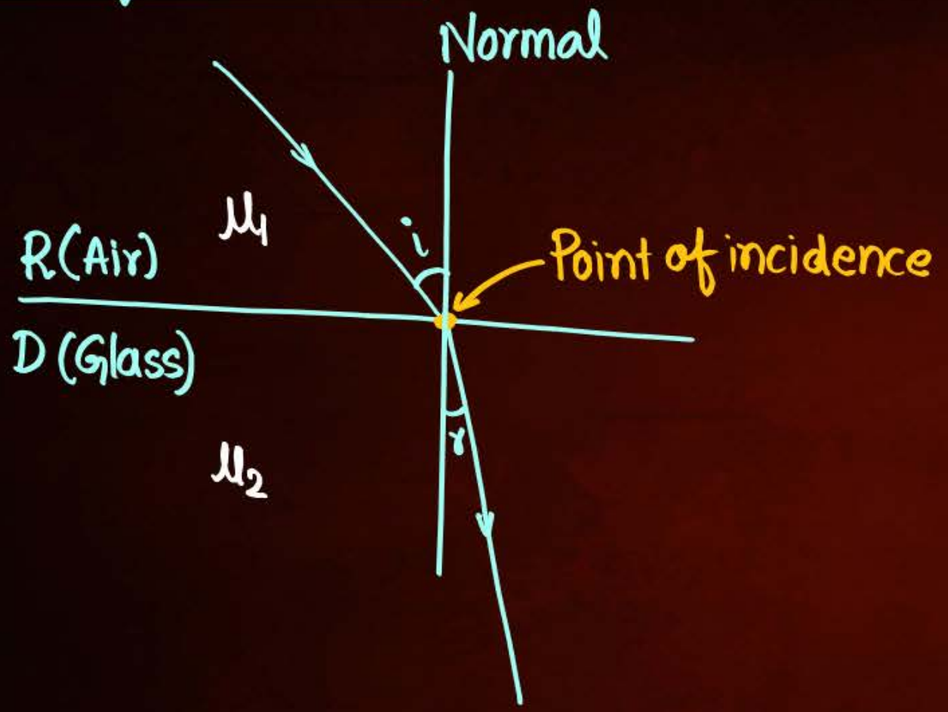
$$\mu_g = \frac{3}{2}$$

$$v_g = \frac{2}{3}c$$

$$= \frac{2}{3} \times 3 \times 10^8$$

$$v_g = 2 \times 10^8 \text{ m/s}$$

Laws of Refraction



① Incident Ray, Normal at point of incidence and refracted ray, all lies in same plane.

② Willebrod Snell's Law

$$\frac{\sin i}{\sin r} = \text{constant}$$

$$\boxed{\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1} = {}_1\mu_2}$$

${}_1\mu_2$ → Refractive Index of 2nd medium w.r.t. 1st medium.

Air ⟶ Glass

$$\frac{\sin i}{\sin r} = \frac{\mu_g}{\mu_a} = {}_a\mu_g$$

Glass ⟶ Water

$$\frac{\sin i}{\sin r} = \frac{\mu_w}{\mu_g} = {}_g\mu_w$$

$$\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1}$$

$$= \frac{f/v_2}{f/v_1}$$

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2}$$

$$= \frac{f\lambda_1}{f\lambda_2}$$

$$\frac{\sin i}{\sin r} = \frac{\lambda_1}{\lambda_2}$$

$v = f\lambda$

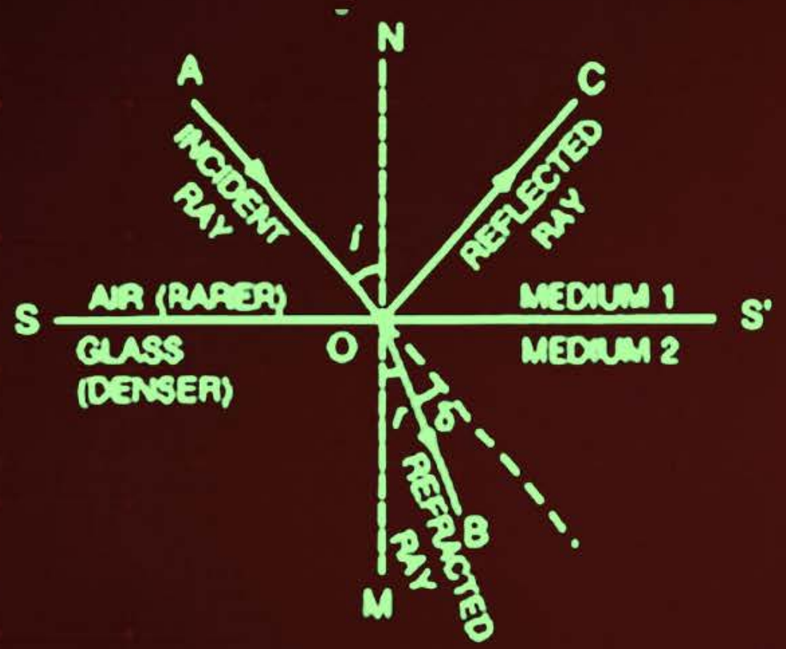
↓

property of source and it doesn't change in refraction or reflection of light.

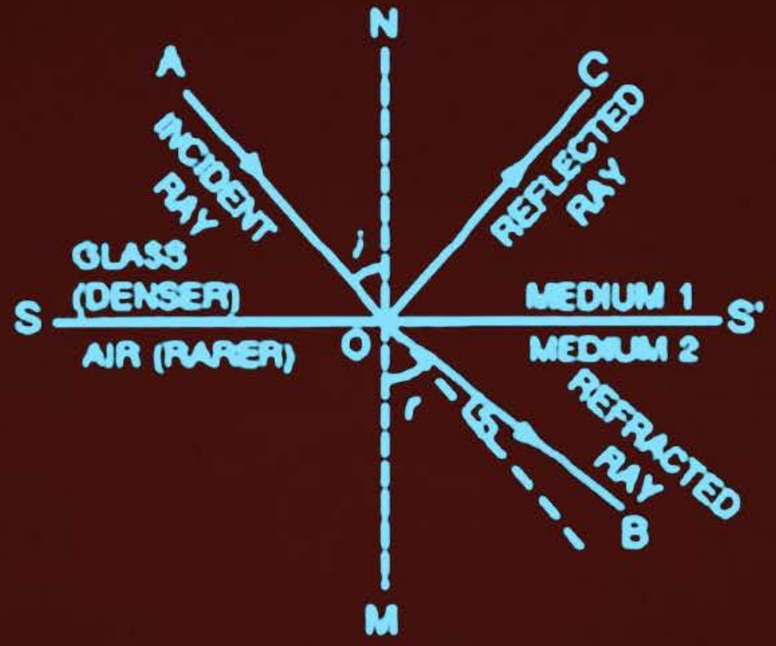
REFRACTION OF LIGHT

The change in direction of the path of light, when it passes from one transparent medium to another transparent medium, is called refraction. The refraction of light is essentially a surface phenomenon.

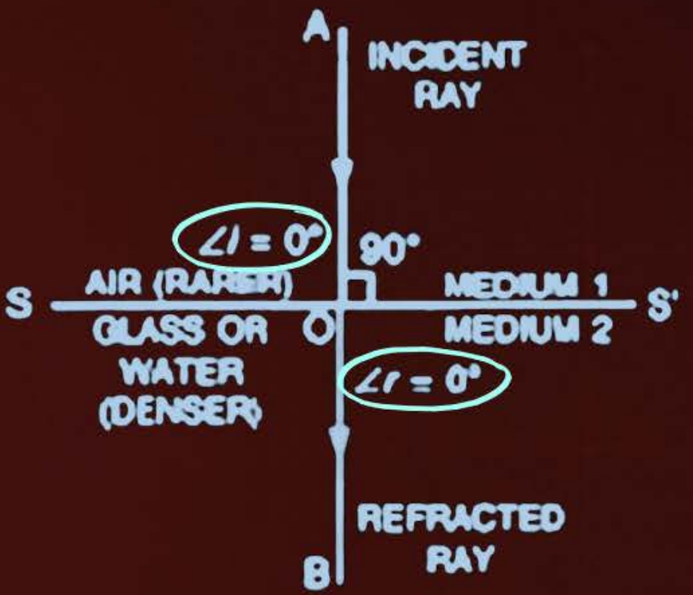
REFRACTION OF LIGHT



Refraction from rarer to denser medium



Refraction from denser to rarer medium

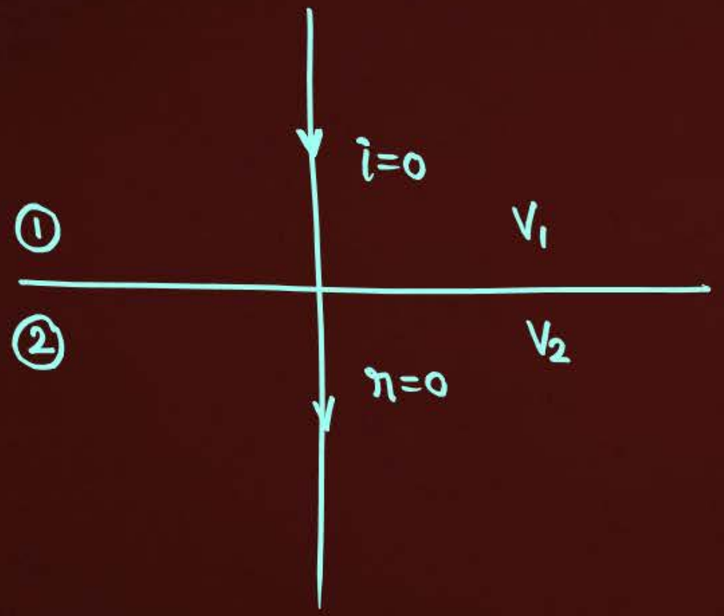


Refraction at normal incidence

CAUSE OF REFRACTION

When a ray of light passes from one medium to another medium, its direction (or path) changes **because of change in speed of light in going from one medium to another**.

- For normal incidence ($\angle i = 0°$), the speed of light changes but the direction of light does not change.



LAWS OF REFRACTION

The refraction of light obeys two laws of refraction which were given by the Dutch scientist **Willebrod Snell**, so they are known as Snell's laws after his name.

(1) The incident ray, the refracted ray and the normal at the point of incidence, all lie in the same plane.

(2) The ratio of the sine of the angle of incidence i to the sine of the angle of refraction r is constant for the pair of given media. i.e., mathematically

$$\frac{\sin i}{\sin r} = \text{constant } {}_1\mu_2$$

The constant is called the refractive index of the second medium with respect to the first medium. It is generally represented by the Greek letter ${}_1\mu_2$ (mew).

REFRACTIVE INDEX

The refractive index of second medium with respect to the first medium is defined as the ratio of the sine of the angle of incidence in the first medium to the sine of the angle of refraction in the second medium.

Unit: The refractive index has no unit as it is the ratio of two similar quantities.

REFRACTIVE INDEX

The refractive index of a medium is defined as the ratio of the speed of light in vacuum (or air) to the speed of light in that medium,

$$\mu = \frac{\text{Speed of light in vacuum or air } (c)}{\text{Speed of light in that medium } (V)}$$

QUESTIONS

The refractive index of water is $4/3$. Explain the meaning of this statement.

Solution:

It means that the speed of light in water is $3/4^{th}$ the speed of light in vacuum (or air). OR the speed of light in air (or vacuum) is 4/3 times the speed of light in water.

QUESTIONS

Refractive index of water is 4/3. Calculate the speed of light in water. Speed of light in vacuum is 3×10^8 ms^{-1}.

$$\mu_w = \frac{4}{3}$$

$$\frac{c}{v_w} = \frac{4}{3}$$

$$v_w = \frac{3}{4}c = \frac{3}{4}\times3\times10^8 = \frac{9}{4}\times10^8$$

$$v_w = 2.25\times10^8 \text{ m/s}$$

QUESTIONS

The refractive index of water is $\frac{4}{3}$ and of glass is $\frac{3}{2}$. What is the refractive index of glass with respect to water?

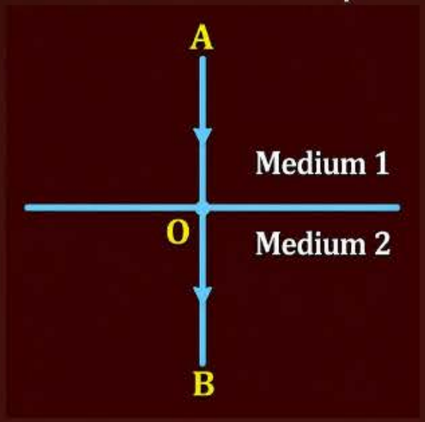
$$\mu_w = \frac{4}{3}$$

$$\mu_g = \frac{3}{2}$$

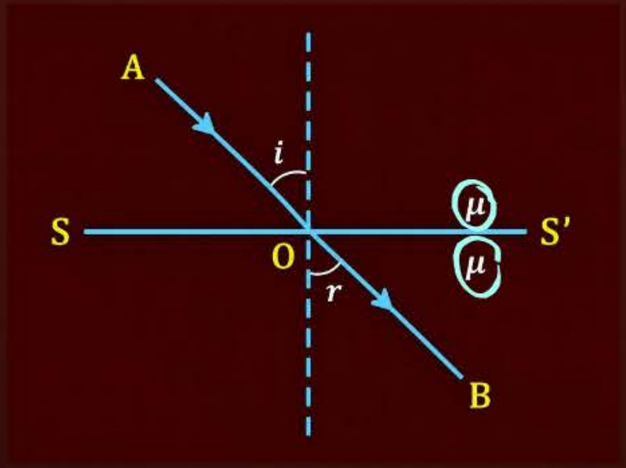
$$w\mu_g = \frac{\mu_g}{\mu_w} = \frac{3/2}{4/3} = \frac{9}{8}$$

CONDITIONS FOR A LIGHT RAY TO PASS UNDEVIATED ON REFRACTION

(1) When the angle of incidence at the boundary of two media is zero ($\angle i = 0°$).



(2) When the refractive index of medium 2 is same as that of medium 1 ($i = r$).



Effect on speed (V), wavelength (λ) and frequency (f) due to refraction of light

$V \rightarrow$ changes

$\lambda \rightarrow$ changes

But $f \rightarrow$ does not change.
(Property of source)

R ⟶ D	
V ↑	V ↓
D ⟶ R	
V ↓	V ↑



λ_1 $\qquad$ λ_2

$V_1 = f\lambda_1$ $\qquad$ $V_2 = f\lambda_2$

$$\frac{V_2}{V_1} = \frac{f\lambda_2}{f\lambda_1}$$

$$\frac{V_2}{V_1} = \frac{\lambda_2}{\lambda_1}$$

$$\lambda_2 = \frac{V_2}{V_1}\lambda_1 \Rightarrow \lambda_2 = \frac{\lambda_1}{{}_1\mu_2}$$

$$\frac{\mu_2}{\mu_1} = \frac{V_1}{V_2}$$

$${}_1\mu_2 = \frac{V_1}{V_2}$$

$$\frac{V_2}{V_1} = \frac{1}{{}_1\mu_2}$$

Effect on speed (V), wavelength (λ) and frequency (f) due to refraction of light

If a ray of light of frequency f and wavelength λ suffers refraction from air (speed of light $= c$) to a medium in which the speed of light is V, then the frequency of light in the medium remains unchanged (equal to f), but the wavelength of light changes to λ'

in air

$$f = \frac{c}{\lambda}$$

in medium

$$f = \frac{V}{\lambda'}$$

$$\therefore \frac{c}{\lambda} = \frac{V}{\lambda'} \qquad \lambda' = \frac{V}{c}\lambda$$

But $\frac{c}{V} = \mu$ the refractive index of the medium.

$$\boxed{\lambda' = \frac{\lambda}{\mu}}$$



Factors affecting the refractive index of a medium

(1) **Nature of the medium (its optical density)** : As smaller the speed of light in a medium relative to air, higher is the refractive index of that medium.

For example

$$V_{\text{glass}} = 2 \times 10^8 ms^{-1}, \quad \mu_{\text{glass}} = 1.5$$

$$V_{\text{water}} = 2.25 \times 10^8 ms^{-1}, \quad \mu_{\text{water}} = 1.33.$$

(2) **Physical condition such as temperature** : With increase in temperature, the speed of light in medium increases, so the refractive index of medium decreases.



Factors affecting the refractive index of a medium

(3) **The colour or wavelength of light** : The speed of light of all colours is same in air (or vacuum), but in any other transparent medium, the speed of light is different for different colours.

- In a given medium, the **speed of red light is maximum** and **that of the violet light is least**,
- therefore the refractive index of that medium is maximum for violet light and least for red light (i.e., $\mu_V > \mu_R$).
- The wavelength of red light is more than that of violet light, so refractive index of a medium decreases with the increase in wavelength.

PRINCIPLE OF REVERSIBILITY OF THE PATH OF LIGHT

According to this principle, the path of a light ray is reversible.

QUESTIONS

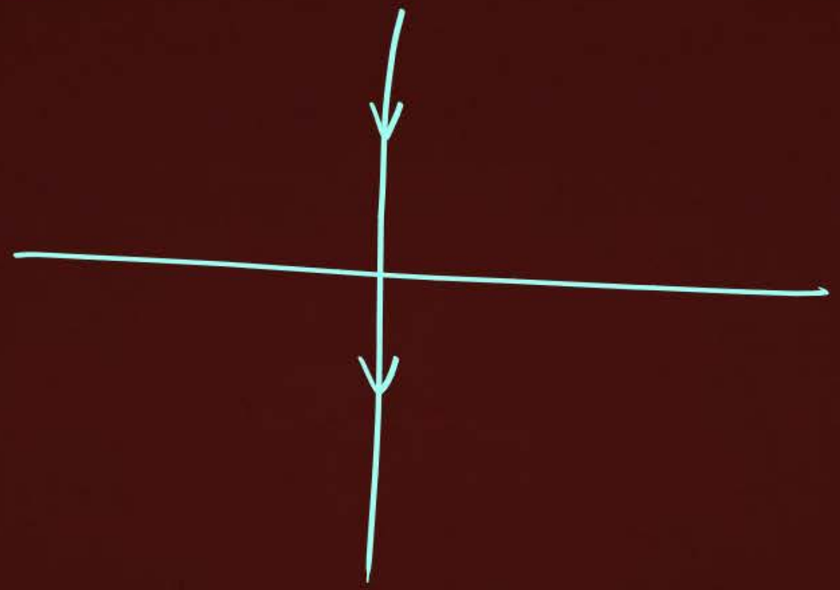
R ⟶ D

When a ray of light from air enters a denser medium, it:

1. bends away from the normal
2. bends towards the normal
3. goes undeviated
4. is reflected back.

QUESTIONS

A light ray does not bend at the boundary in passing from one medium to the other medium if the angle of incidence is:



1. 0°
2. 45°
3. 60°
4. 90°

QUESTIONS

The highest refractive index is of

1. Glass
2. Water
3. Diamond (2.42)
4. Ruby

QUESTIONS

Given the following mediums, in which medium does light travel the fastest?

1. Air
2. Water
3. Glass
4. Wood

QUESTIONS

Figure shows a ray of light as it travels from medium (A) to medium (B). Refractive index of the

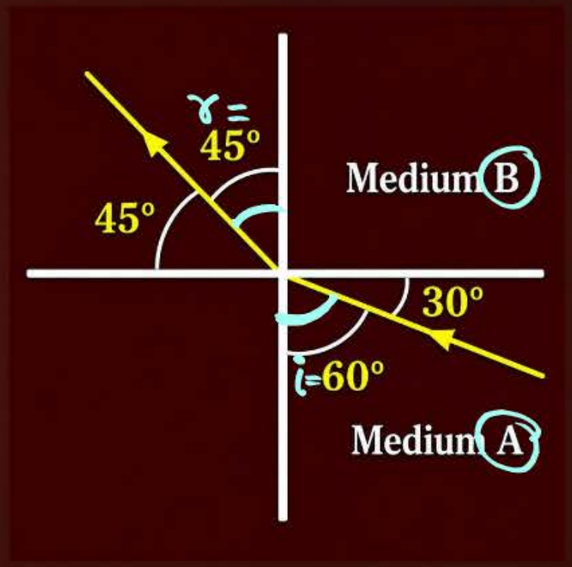
1. $\sqrt{3/2}$
2. $\sqrt{2/3}$
3. $\sqrt{1/2}$
4. $\sqrt{2}$

$$\frac{\sin i}{\sin r} = \frac{\mu_B}{\mu_A}$$

$$_A\mu_B = \frac{\sin 60^\circ}{\sin 45^\circ} = \frac{\sqrt{3}/2}{1/\sqrt{2}}$$

$$= \frac{\sqrt{3}\cdot\sqrt{2}}{2}$$

$$= \frac{\sqrt{3}}{\sqrt{2}}$$



QUESTIONS

Imp Orange light of wavelength 6600Å travelling in air gets refracted in water. If the speed of light in air is 3×10^8 ms^{-1} and refractive index of water is $4/3$, find :

(i) the frequency of light in air,

(ii) the speed of light in water, and

(iii) the wavelength of light in water.

Air ⟶ Water

$\lambda_1 = 6600\,A^\circ$

$V_1 = 3\times10^8$ m/s

$\mu_W = \frac{4}{3} \Rightarrow V_W = \frac{3}{4}\times3\times10^8 = \frac{9}{4}\times10^8 = 2.25\times10^8$ m/s.

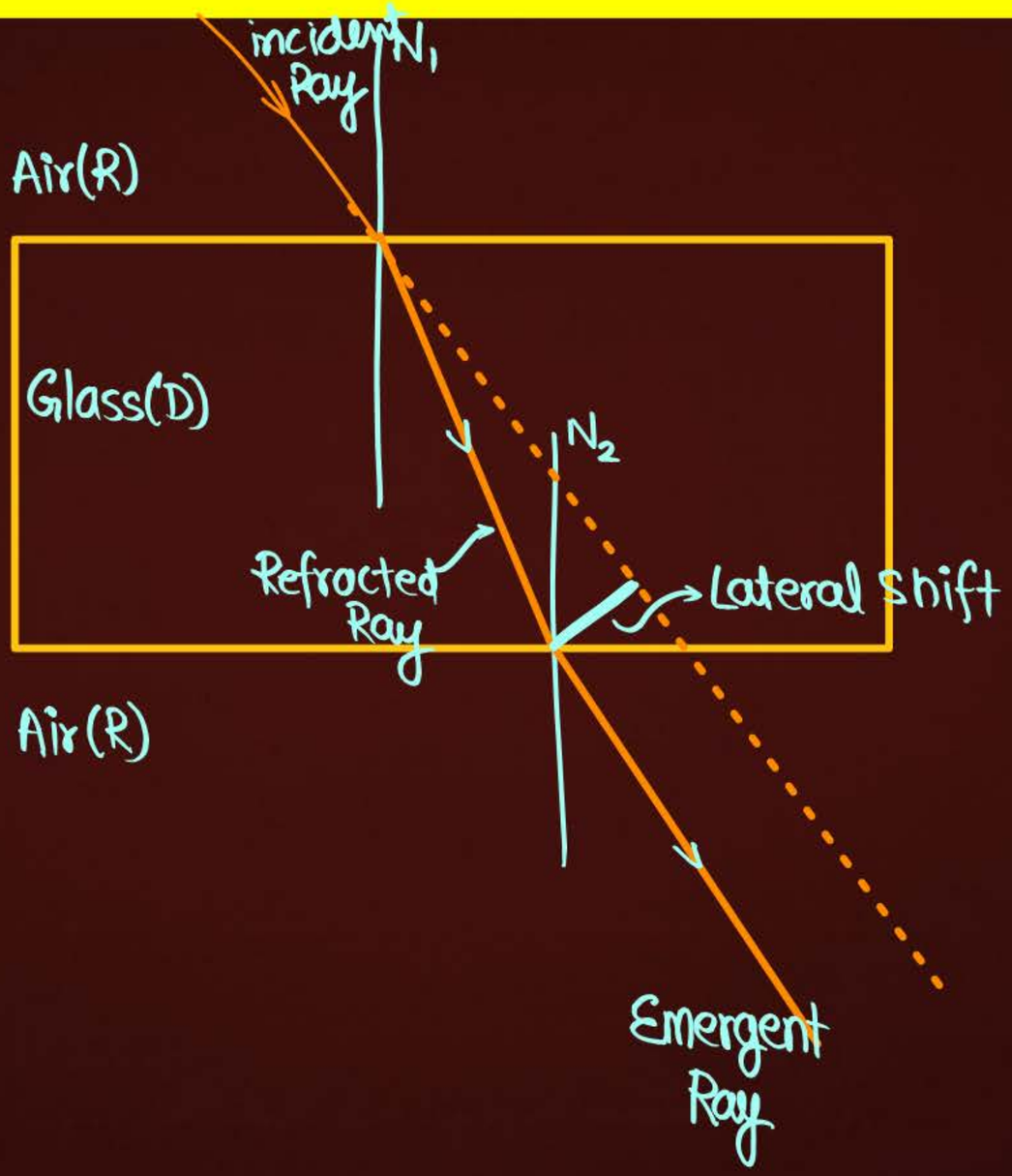
$\lambda_2 = \frac{\lambda_1}{{}_1\mu_2} = \frac{6600\,A^\circ}{\left(\frac{4}{3}\right)} = 4950\,A^\circ$

$\boxed{V_1 = f\lambda_1}$

$3\times10^8 = f(6600\times10^{-10})$

$f = \frac{3\times10^8}{6600\times10^{-10}} = \frac{3\times10^{16}}{66} = \frac{300}{66}\times10^{14} = 4.54\times10^{14}$ Hz.

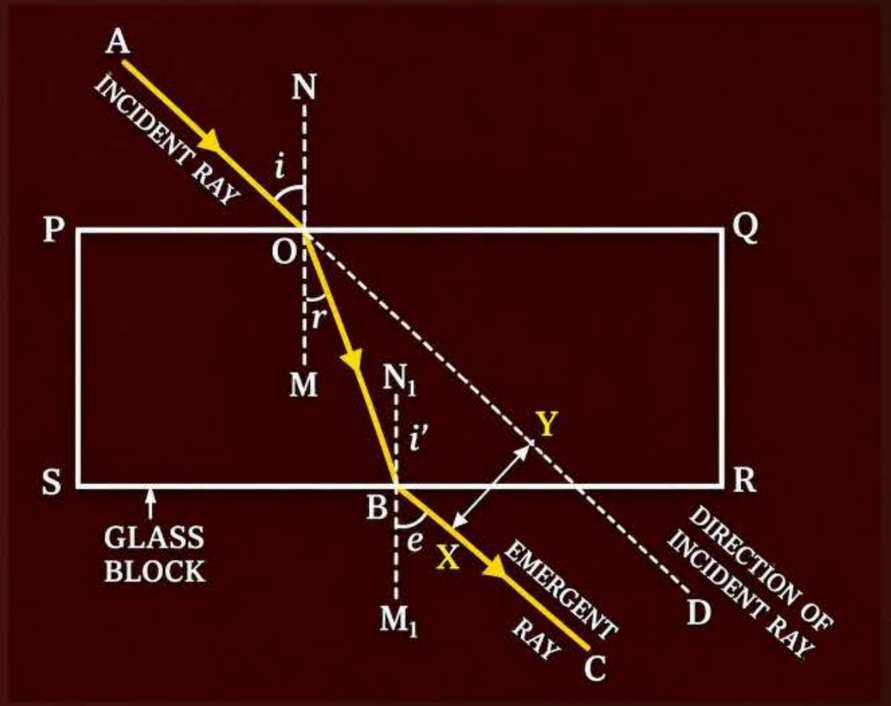
REFRACTION OF LIGHT THROUGH A RECTANGULAR GLASS BLOCK



REFRACTION OF LIGHT THROUGH A RECTANGULAR GLASS BLOCK

Lateral Displacement

- Due to refraction of light at two parallel surfaces of a parallel sided glass block, the angle of incidence is equal to the angle of emergence, so the incident ray AO and the emergent ray BC are parallel (i.e., they are in same direction), but they are not along the same line.
- The emergent ray is laterally displaced from the path of incident ray. The path of incident ray AO in absence of glass block has been shown by the dotted line OD.
- The perpendicular distance XY $(= x)$ between the path of emergent ray BC and the direction of incident ray OD is called the lateral displacement.



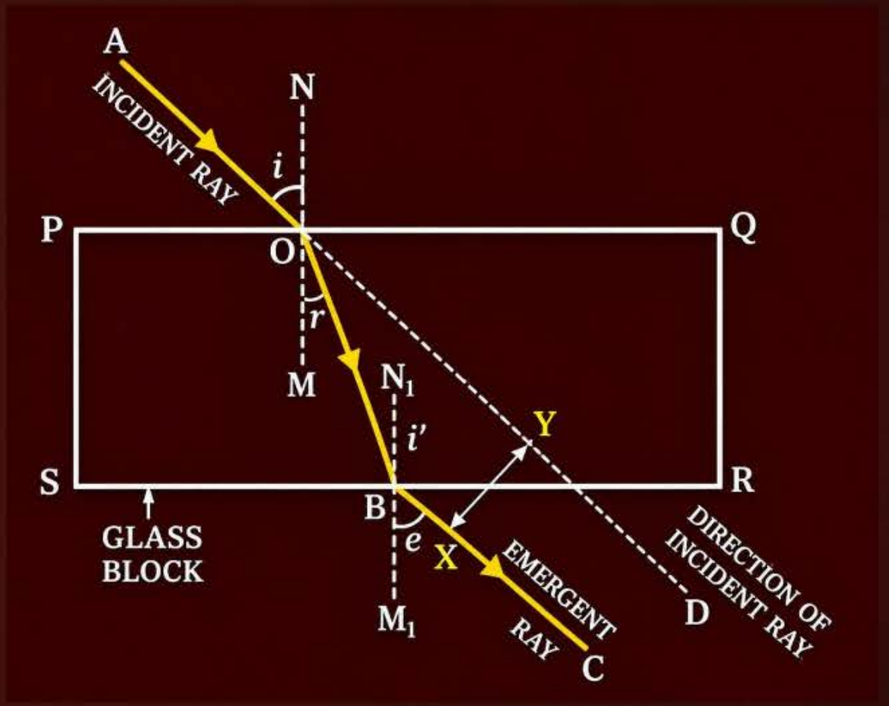
REFRACTION OF LIGHT THROUGH A RECTANGULAR GLASS BLOCK

Lateral Displacement

(i) **Dependence on the thickness of medium**: More the thickness of the medium, more is the lateral displacement.

(ii) **Dependence on the angle of incidence** : More the angle of incidence, more is the lateral displacement.

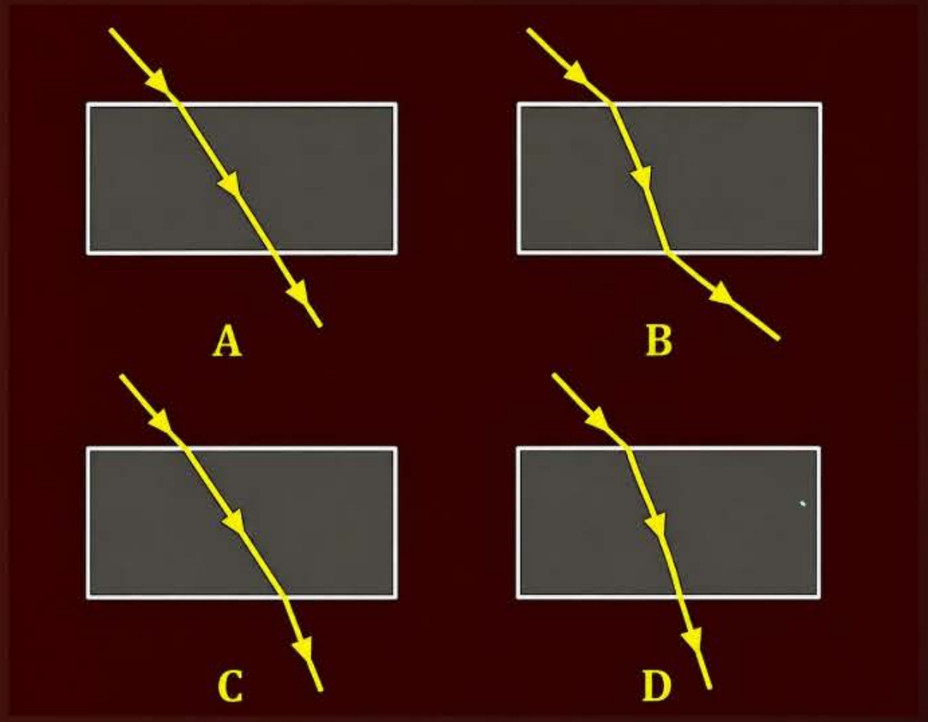
(iii) **Dependence on the refractive index** : More the refractive index of the medium, more is the lateral displacement. Since refractive index increases with the decrease in wavelength of light, so the lateral displacement increases with the decrease in wavelength of light (i.e., lateral displacement is more for violet light than for red light).

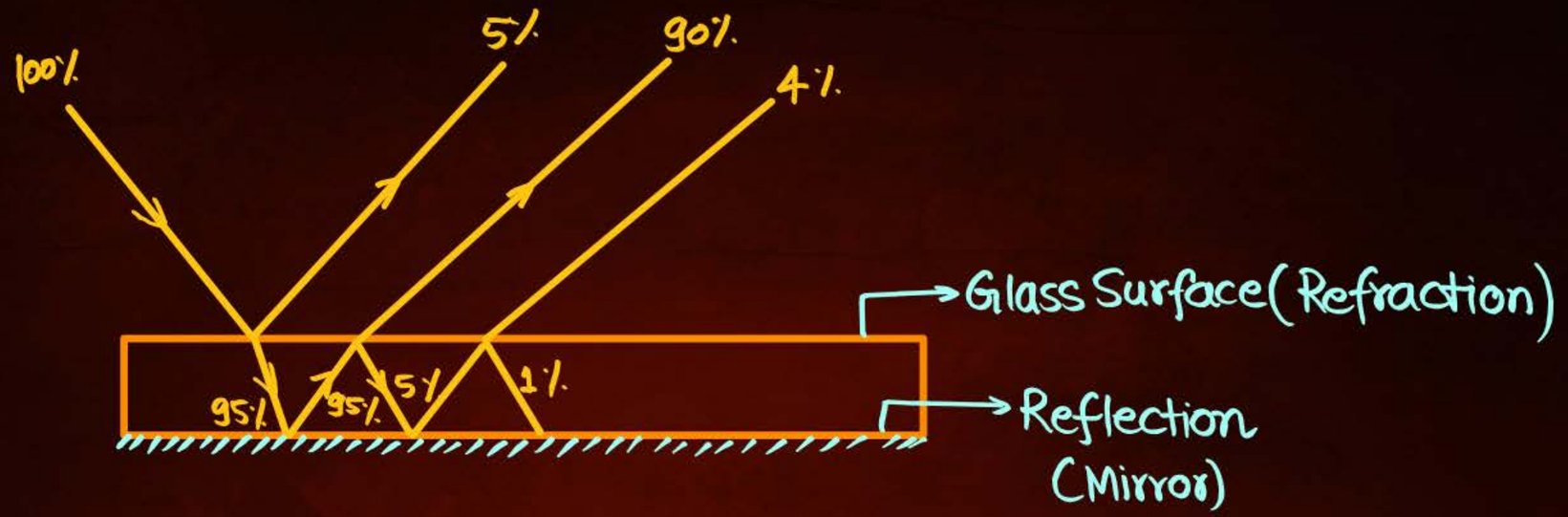


QUESTIONS

The path of a ray of light coming from air passing through a rectangular glass slab traced by four students are shown as A, B, C and D in fig. Which one of them is correct?

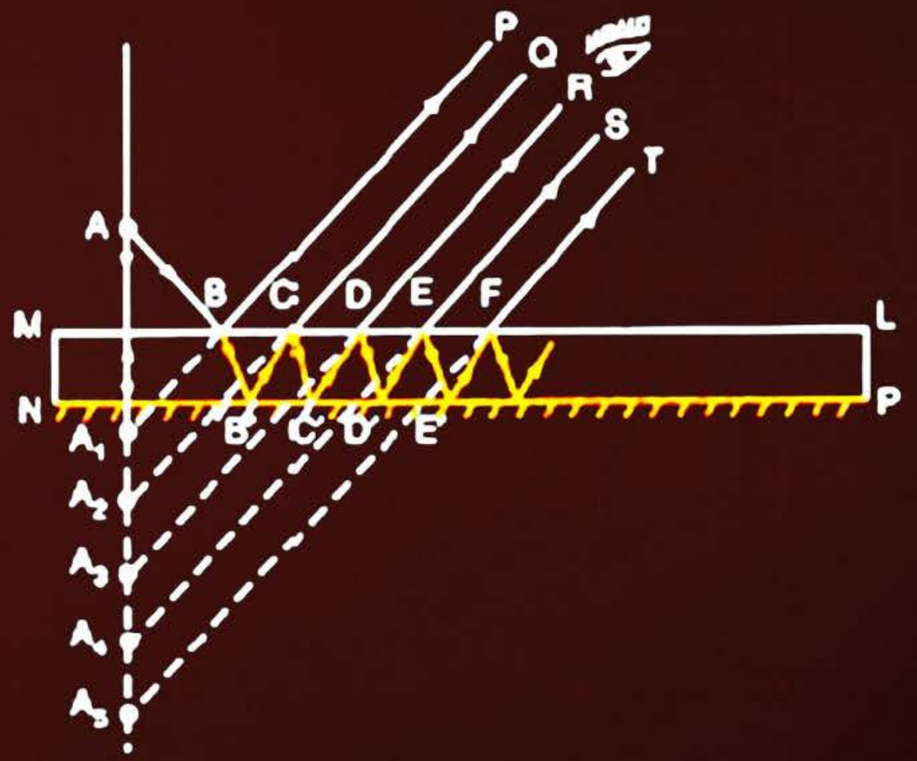
1. A
2. B
3. C
4. D




100%
5%
90%
4%
95%
95%
5%
1%
Glass Surface (Refraction)
Reflection
(Mirror)

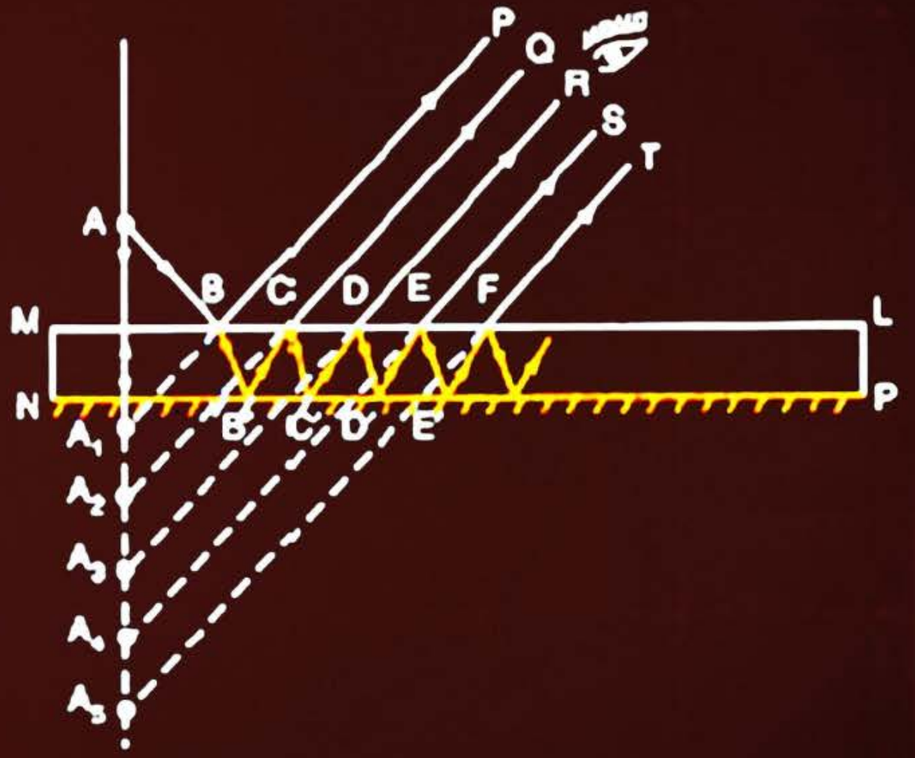
MULTIPLE IMAGES IN A THICK PLANE GLASS PLATE OR THICK MIRROR

If an object is placed in front of a thick plane glass plate (or a thick mirror) and is viewed obliquely, a number of images are seen. Out of these images, the **second image is the brightest**, while others are of decreasing brightness.



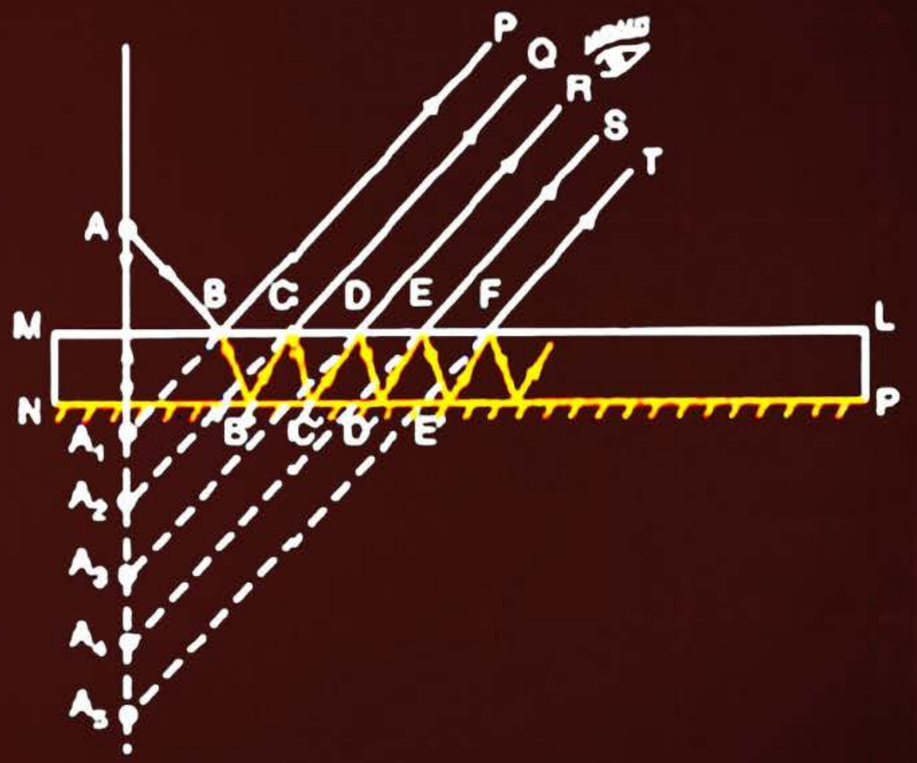
MULTIPLE IMAGES IN A THICK PLANE GLASS PLATE OR THICK MIRROR

- Consider two rays, one falling normally on the mirror and the other AB falling obliquely on it.
- When the ray of light AB falls on the surface LM of the mirror, a small part of light (nearly 4%) is reflected in the direction BP, forming a faint virtual image at A_1,
- while a larger part of light (nearly 96%) is refracted along BB' inside the glass. The ray BB' which strikes at B', is now strongly reflected back by the silvered surface PN inside glass as $B'C$. This ray is then partially refracted along CQ in air and partially reflected along CC' within the glass. The refracted ray CQ forms the virtual image A_2.
- The image A_2 is the brightest image because it is due to the light suffering a strong first reflection at the silvered surface PN.



MULTIPLE IMAGES IN A THICK PLANE GLASS PLATE OR THICK MIRROR

- The reflected ray CC' further suffers multiple reflections at $C', D, D', \ldots$ and refractions at $D, E, F, \ldots \ldots$ within the thickness of glass plate giving rise to multiple virtual images $A_3, A_4, A_5, \ldots\ldots$ of gradually decreasing brightness.



HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

QUESTIONS

A ray of light in passing from one transparent medium to the other medium having different optical density, bends.

(a) Name the phenomenon. Give reason for it.

(b) How do the following quantities change: speed, wavelength, frequency and amplitude if second medium is denser than the first medium.

(c) State whether the ray of light will bend or not, if both medium have same optical densities.

QUESTIONS

- The speed of light in air is $3 \times 10^8 \text{ms}^{-1}$. Calculate the speed of light in glass. The refractive index of glass is 1.5 .

[Ans. $2 \times 10^8 \text{ms}^{-1}$]

- The speed of light in diamond is $125{,}000 \text{kms}^{-1}$. What is its refractive index ? (Speed of light in air $= 3 \times 10^8 \text{ms}^{-1}$).

[Ans. 2.4]

- The refractive index of water with respect to air is $4/3$. What is the refractive index of air with respect to water?

[Ans. 0.75]

- A ray of light of wavelength 5400Å suffers refraction from air to glass. Taking ${}_a\mu_g = 3/2$, find the wavelength of light in glass.

[Ans. 3600Å]



Thank You